

ENTERPRISE SYSTEM ARCHITECTURE & TECHNICAL SPECIFICATION DOCUMENT

Project: Centralized AI-Enabled Real Estate CRM Engine

Author: Principal Lead Solutions Architect

Client: Kohinoor Group Pune, India

Stack: FastAPI, Next.js, Redis, Celery, PostgreSQL

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Section 1: Executive Architecture Overview

This system specification details the core technical blueprint for the next-generation centralized, AI-enabled Customer Relationship Management (CRM) platform commissioned for the **Kohinoor Group**. The architectural mission is to unify high-velocity marketing pipelines, conversational sales engines, multi-site interactive unit locking systems, finance-linked accounts structures, and long-term customer satisfaction matrices ('Sada Sukhi Raho' ecosystem) into an integrated low-latency platform.

1.1 Operational Concurrency Constraints

Real estate development workflows present unique technical anomalies. Unlike micro-transaction retail e-commerce, property allocation processes demand long-lived transactional integrity backed by atomic validation rules. When premium residential projects go live, marketing automation funnels inject thousands of unstructured data vectors into our systems via third-party webhooks. At the exact same instant, hundreds of localized site sales agents query the physical tower matrix grids via client webapps while buyers sit at their tables making token choices.

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Section 2: Technical Stack Selection & Justification

The core stack has been rigorously engineered around high-performance parameters, low context-switching latency, and robust structural safety under heavy operational stress. The API framework uses FastAPI over legacy frameworks due to native async routing handling loops. Traditional web servers block resources per user, which degrades during launch phases.

2.1 Async Framework Advantages

FastAPI operates natively via the ASGI standard, processing incoming requests within an asynchronous event-loop. Connections remain open as non-blocking tasks while underlying tasks await I/O returns from databases or caches.

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Section 3: End-to-End Data Ingestion Pipelines

The data lifecycle follows a strictly choreographed pipeline that moves through ingestion, background grading, distribution, state mutation, financial balancing, and management aggregation.

3.1 Webhook Endpoint Structure

Webhooks stream parameters securely from listing networks into an async task orchestrator pool to return instant network updates without holding main threads open.

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Section 4: Module 1 & 2 - Marketing Automation & Sales

The Marketing Module isolates primary transactional engines from heavy external promotional pools. Leads are parsed through localized AI processing clusters to gauge customer preference metrics.

4.1 Conversation Schedulers

The Sales Engine leverages LangChain layers to parse loose text entries from tools like WhatsApp Business API, managing calendar availability and setting local site agendas seamlessly.

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Section 5: Module 3 - Real-Time Inventory Locking

The Inventory Module manages real-time status matrices of building wings, levels, and layouts across distributed sites without risking data collision states.

5.1 Atomic Exclusion Patterns

To prevent double-bookings, the gateway enforces atomic Redis holds using key exclusions, circumventing slow persistent transactional locks across nodes.

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Section 6: Module 4 & 5 - Accounts & Support Lifecycle

The Finance block links transactions directly to the physical milestones managed by engineering monitors (e.g. foundational completions or floor structures).

6.1 Sada Sukhi Raho Systems

The Customer Service system transitions users cleanly post-sale, tracking paperwork workflows and verifying KYC uploads securely using integrated AI OCR processors.

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Section 7: Module 6 & 7 - Channel Partners & Management

Brokers query property metrics securely through isolation wrappers. Cryptographic phone checks ensure attribution remains accurate and prevents payout overlapping.

7.1 Executive Insights OLAP

Strategic modules tap into asynchronous database read replicas, providing clean management dashboard visualizations without locking real-time site transactional systems.

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Section 8: Module 8 - System Security & RBAC Guardrails

Data access lines are restricted using strict Role-Based Access Control filters injected directly into framework dependencies.

8.1 Relational Blueprint Schemas

Database layers implement structured data patterns to ensure tracking safety and full audit logging capacity across execution lifetimes.

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Section 9: Advanced Database Schema Specifications

Ledger definitions map booking steps against property matrices to maintain balanced record trails across distributed cluster installations.

9.1 Financial Audit Tracking Tables

Billing tables log demanded amounts, tax parameters, and due milestones with microsecond accuracy to support internal audit streams.

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Section 10: System Integration Matrices

Outbound platform bridges use idempotent retry loops to maintain sync stability even during external network drop windows.

10.1 Concurrency Load Controls

Quality frameworks utilize automated network emulation loops to simulate hundreds of concurrent agent requests hitting the primary allocation APIs.

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Section 11: Enterprise Cloud Infrastructure Layout

The deployment layout leverages Amazon Web Services (AWS) Mumbai region instances inside secured Virtual Private Clouds to preserve complete local data residency compliance.

11.1 Disaster Recovery Protections

Zero-data-loss guidelines dictate continuous storage streams, allowing recovery states to clear within granular intervals in case of physical node interruptions.

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